Cross-Query Score Comparability in Sparse Pairwise Learning: Residual Centering with Uncertainty-Aware Shrinkage

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Abstract

In sparse pairwise prediction, candidates from multiple queries are often prioritized under a shared screening budget, requiring scores to be comparable across queries rather than merely well-ranked within individual queries. However, auxiliary information fusion may improve within-query discrimination while introducing systematic query-level score shifts, potentially compromising global candidate prioritization. To address this issue, we propose Query-Level Residual Centering (QRC), a lightweight post-hoc correction method that estimates and removes auxiliary-induced common shifts using matched Base and Raw predictions while exactly preserving within-query score differences. Furthermore, Shrinkage QRC (SQRC) adaptively adjusts the correction strength according to estimation uncertainty, mitigating potential overcorrection. Theoretical analysis establishes ranking invariance and exact recovery under additive nuisance shifts, while controlled experiments characterize the conditions under which correction is beneficial. Extensive experiments on miRNA–drug association prediction demonstrate that QRC improves pooled AUROC by 5.34 and 4.74 percentage points under drug-cold and strict-cold evaluation, respectively. Cross-backbone experiments and independent validation using PRISM phenotypic information further demonstrate the reproducibility of the observed score-comparability phenomenon. Under estimation stress, SQRC achieves lower bias-estimation mean squared error in 563 of 800 cases. Conversely, drug–target interaction experiments reveal that query-level shifts may contain task-relevant information, highlighting the limitations of indiscriminate correction. Overall, QRC and SQRC provide model-agnostic, conditionally applicable correction strategies for improving cross-query score comparability in sparse pairwise learning.

Keywords: Reliable machine learning, Multimodal information fusion, Query-level residual centering, Cross-query score comparability, Intelligent decision support

1. Introduction

Machine learning models are increasingly deployed as components of decision-making systems rather than as isolated predictors. In sparse pairwise learning, a model estimates the relevance or interaction strength between a query and a set of candidates, allowing multiple queries to compete for limited resources through a global ranking, shared screening budget, or common decision threshold. Such scenarios arise in biomedical screening, recommendation, and other candidate prioritization tasks [1]. A recent hypergraph sparse-transformer study illustrates miRNA–drug sensitivity prediction [2]. For example, in miRNA–drug association prediction, multiple unseen drugs may be evaluated against thousands of candidate miRNAs, while only a limited number of predicted associations can be experimentally validated. Although conventional learning objectives primarily emphasize predictive discrimination, the effectiveness of these downstream decisions also depends on whether scores generated for different queries are mutually comparable.

A fundamental difficulty is that accurate ranking within individual queries does not necessarily imply reliable ranking across queries. Consider a model that assigns scores to all candidates associated with a particular query. Adding a constant offset to these scores leaves every within-query ordering unchanged, yet alters their positions relative to candidates belonging to other queries. Consequently, an artificially elevated query may dominate a global Top-N list or consume a disproportionate share of a shared screening budget, even when its internal candidate ranking remains accurate. This reveals an important distinction between within-query discrimination and cross-query score comparability, two properties that are often conflated when predictive performance is evaluated using conventional ranking metrics.

This issue becomes particularly relevant when auxiliary information is incorporated into sparse learning problems. Because observed interactions cover only a small fraction of the complete query–candidate space, modern prediction models frequently integrate structural features, sequence representations, phenotypic profiles, or pretrained embeddings to enrich otherwise limited supervision. Although such information can improve representation quality and candidate discrimination, its influence on the output score space is not necessarily uniform.

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Specifically, auxiliary fusion may produce candidate-specific score changes that affect relative rankings, together with query-level common shifts that translate an entire group of predictions. The former may improve local discrimination, whereas the latter can alter global candidate allocation without changing local rankings. Foundation-model transfer has also been used to predict drug responses in previously unseen cell types [3]. Improving feature representations therefore does not automatically guarantee reliable cross-query decisions.

The distinction can be examined through two complementary evaluation perspectives. Query-macro metrics, such as query-wise AUROC and AUPR averaged across queries, measure discrimination within individual candidate sets and are invariant to query-specific additive score shifts. In contrast, pooled metrics evaluate all query–candidate pairs in a shared score space and are sensitive to the relative locations of query-specific score distributions. Therefore, auxiliary fusion may improve query-macro performance while simultaneously degrading pooled performance. Such disagreement suggests a potential score-placement problem rather than necessarily indicating inferior feature representations or candidate discrimination. More importantly, conventional global score normalization does not explicitly isolate shifts introduced by auxiliary information, while direct query-wise centering may also remove query-level information already captured by the original predictor.

Motivated by this observation, we propose Query-Level Residual Centering (QRC), a lightweight, model-agnostic post-hoc correction method for improving cross-query score comparability. QRC operates on matched predictions from a Base model without the target auxiliary source and a Raw model incorporating that source under otherwise comparable configurations. By decomposing the Raw-minus-Base score increment into a query-level common component and candidate-specific residuals, QRC estimates and removes the auxiliary-induced common shift while retaining the candidate-specific contribution. Unlike direct score centering, the proposed correction preserves the query-level location information already present in Base. Moreover, subtracting a common offset within each query guarantees that all within-query score differences and rankings remain unchanged.

A remaining challenge is that the estimated query-level shift may be unreliable when candidate sets are small or score increments are highly variable. To address this limitation, we further develop Shrinkage QRC (SQRC), which introduces an uncertainty-aware random-effects shrinkage mechanism to adaptively regulate correction strength. Rather than imposing full centering on every query, SQRC attenuates corrections associated with uncertain shift estimates. We establish within-query ranking invariance, query-macro metric invariance, and exact recovery under a controlled additive nuisance model. Importantly, these properties do not imply that removing every query-level shift is beneficial: common shifts may also contain task-relevant information, making their correction inherently conditional.

We investigate this mechanism through controlled experiments and complementary real-world evaluations. In miRNA–drug association prediction, QRC improves pooled AUROC by 5.34 and 4.74 percentage points under drug-cold and strict-cold evaluation, respectively, without changing within-query rankings. Additional experiments involving different prediction backbones and an independent PRISM auxiliary source examine whether the phenomenon persists beyond a single model configuration. Uncertainty stress tests evaluate the benefits of SQRC, while drug–target interaction experiments on Davis reveal an important boundary case in which retaining query-level information is advantageous. Together, these results position QRC and SQRC as conditional reliability mechanisms for machine learning systems that support joint candidate prioritization.

The main contributions of this work are summarized as follows:

- **Problem formulation and reliability analysis.** We identify auxiliary-induced query-level score shifts as a distinct reliability problem in sparse pairwise learning and clarify why improved within-query discrimination does not necessarily translate into reliable cross-query candidate prioritization. A matched-prediction formulation separates query-common shifts from candidate-specific score changes.

- **Model-agnostic correction with theoretical properties.** We propose QRC to estimate and remove auxiliary-induced query-level shifts while exactly preserving within-query rankings, and extend it to SQRC through uncertainty-aware shrinkage. We establish ranking and query-macro invariance, analyze exact recovery under additive nuisance shifts, and characterize conditions under which correction may be inappropriate.

- **Comprehensive validation and applicability boundaries.** We construct a staged evaluation covering controlled perturbations, cold-start miRNA–drug association prediction, cross-backbone experiments, independent PRISM auxiliary information, uncertainty stress tests, and cross-task drug–target interaction prediction. These investigations evaluate correction effectiveness, transferability, robustness, and the boundary between nuisance shifts and informative query-level signals.

2. Related Work

2.1. Sparse Pairwise Learning and Auxiliary Information Fusion

Sparse pairwise learning combines interaction-based and feature-based predictors. For heterogeneous graphs, event-aware node representations support link prediction [4], while explicit context extraction helps aggregate relevant neighborhoods [5]. Variational graph autoencoders with pairwise learning have been applied to sparse links [6], and graph masked autoencoders address multi-scale link evidence [7]. Other formulations use ranking-triplet adversarial objectives [8] or transfer links across knowledge graphs [9]. These methods improve

representation and relational inference, but do not directly test whether scores from different query groups occupy a common scale.

In molecular interaction learning, multi-view feature fusion has been used for drug–target prediction [10]. Hierarchical meta-path transformers [11] and motif-based graph transformers [12] exploit structured relational context. Multi-view heterogeneous graph autoencoders also extend this strategy to protein–protein interaction prediction [13]. Cross-modality attention [14], multi-view contrastive learning [15], and equivariant graph encoders with multi-level contrastive objectives [16] offer further ways to represent drug and target features. These approaches focus on estimating interactions from richer structure; our question concerns the comparability of their outputs across queries.

Recent multimodal DTI models combine language-model embeddings with dynamic graph aggregation [17], hierarchical local-to-global attention [18], and geometric graph networks [19]. Joint prediction of drug–target and drug–disease relations expands the supervision available for drug repositioning [20]. Multi-domain heterogeneous networks [21] and twin cross-contrastive fusion [22] provide complementary representations. Fusion generally occurs before prediction; none of these architectural choices alone guarantees that scores for different unseen drugs share a comparable location.

Other recent DTI studies investigate Kolmogorov–Arnold feature fusion [23], joint sequence–structure modeling [24], and self-supervised structural embeddings [25]. Metapath-driven transformers have also been developed for fine-grained metabolite–drug interactions [26]. In adjacent molecular tasks, pseudo-force-field meta-learning supports three-dimensional property prediction [27], whereas cluster-aware disentanglement addresses few-shot properties [28]. These results motivate careful evaluation of new auxiliary sources, especially under entity-cold splits. We therefore study score changes after fusion rather than redesigning the encoders themselves.

2.2. Score Reliability and Cross-Query Comparability

Generalization under distribution shift is a separate but related reliability challenge. Graph studies use causal representation learning [29], contrastive objectives [30], and models that evolve under non-stationary environments [31]. Invariant information learning across environments [32] and Siamese graph encoders [33] further target representations that transfer outside training conditions. Sinkhorn-attention subgraph extraction attempts to identify more stable graph structure [34]. In molecular settings, context-aware codebooks support out-of-distribution property prediction [35], while external knowledge and adaptive prompts have been used for OOD detection [36]. These approaches address representation shift; they do not isolate the location of a score block induced by one auxiliary source.

Continual test-time training with adaptive graph prompts [37] is another response to distribution shift. In our setting, however, the predictors are frozen and correction is applied only to their matched outputs. Entity-cold evaluation can reveal a pooled-performance decline, but pooled metrics alone do not distinguish degraded within-query ranking from cross-query score misplacement.

Probability calibration is related but not identical. It asks whether predicted probabilities align with empirical frequencies, whereas our target is the relative location of score distributions across queries. Online knowledge-distillation research has considered calibration of aggregated predictions [38]; open-set transfer research has addressed overconfidence on unseen samples [39]. Neither objective alone identifies the query-common component of the Raw-minus-Base increment. A global monotone probability map may improve probability quality while leaving heterogeneous query offsets unresolved.

2.3. Centering, Normalization, and Post-hoc Correction

Direct query-wise centering and z-score normalization can align score distributions, but they remove both newly introduced shifts and pre-existing query differences that may already be encoded by the baseline model. Global residual centering has the opposite limitation: one global offset cannot correct heterogeneous query-specific movement. QRC uses a matched difference, Raw minus Base, so the correction is restricted to the component that appears after the target auxiliary source is added. This requirement is restrictive, yet it gives the procedure a clear estimand and avoids redefining the entire score scale.

The matched-difference view also makes the method falsifiable. Additive query shifts are only one possible output distortion. Query-specific rescaling, nonlinear warping, or deterioration of within-query ranking violates the basic model. A common shift can also be informative. We therefore treat QRC as a conditional correction with explicit diagnostic and experimental boundaries, rather than as a universal post-processing step.

3. Methodology

3.1. Problem Formulation

Let q_i denote query i and c_j the jth candidate paired with that query. For pair (i, j), S_{ij}^{raw} and S_{ij}^{base} denote the scores produced by matched Raw and Base models, respectively. Raw introduces the target auxiliary source, while the remaining architecture, data split, loss, and evaluation protocol are kept as closely matched as possible. We define the auxiliary-induced score increment by

$$\Delta_{ij} = S_{ij}^{\text{raw}} - S_{ij}^{\text{base}}, \tag{1}$$

Here Δ_{ij} is the matched score change associated with adding the target auxiliary source. It is a model-comparison quantity, rather than a causal effect identified from the observed data. We decompose this increment into

$$\Delta_{ij} = b_i + r_{ij}, \tag{2}$$

where b_i is the additive location component shared by all candidates of query i, whereas r_{ij} is the candidate-specific residual. Only r_{ij} can alter the relative order of candidates within a fixed query. This is a statistical decomposition: we assign no causal biological meaning to r_{ij} and do not presume that b_i is nuisance.

3.2. *Query-Level Residual Centering*

For query i, let n_i be the number of available candidates. QRC estimates the common location term by the mean auxiliary-induced increment, denoted $\widehat{b}_i$:

$$\widehat{b}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} \Delta_{ij}. \tag{3}$$

The QRC-corrected score subtracts $\widehat{b}_i$ from the Raw prediction:

$$S_{ij}^{\mathrm{QRC}} = S_{ij}^{\mathrm{raw}} - \widehat{b}_i. \tag{4}$$

The use of a matched difference is essential. Centering Raw scores directly would remove the entire Raw query mean, including query-level structure already present in Base. QRC removes only the average score change associated with the added source. The estimator has no trainable parameters and does not require retraining after Base and Raw predictions have been generated.

3.3. *Invariance and Additive Recovery*

For any two candidates j and k attached to the same query, subtracting a common query-specific constant yields

$$S_{ij}^{\mathrm{QRC}} - S_{ik}^{\mathrm{QRC}} = S_{ij}^{\mathrm{raw}} - S_{ik}^{\mathrm{raw}}. \tag{5}$$

Within-query ranking is therefore identical before and after QRC. Any ranking metric computed separately inside each query, including query-wise AUROC and AUPR, is unchanged; averaging these values over queries remains unchanged as well. A change in pooled performance after QRC is consequently attributable to the relative placement of different queries in the shared score space.

A controlled nuisance model makes the recovery target explicit. Suppose Raw scores are generated from Base scores, a

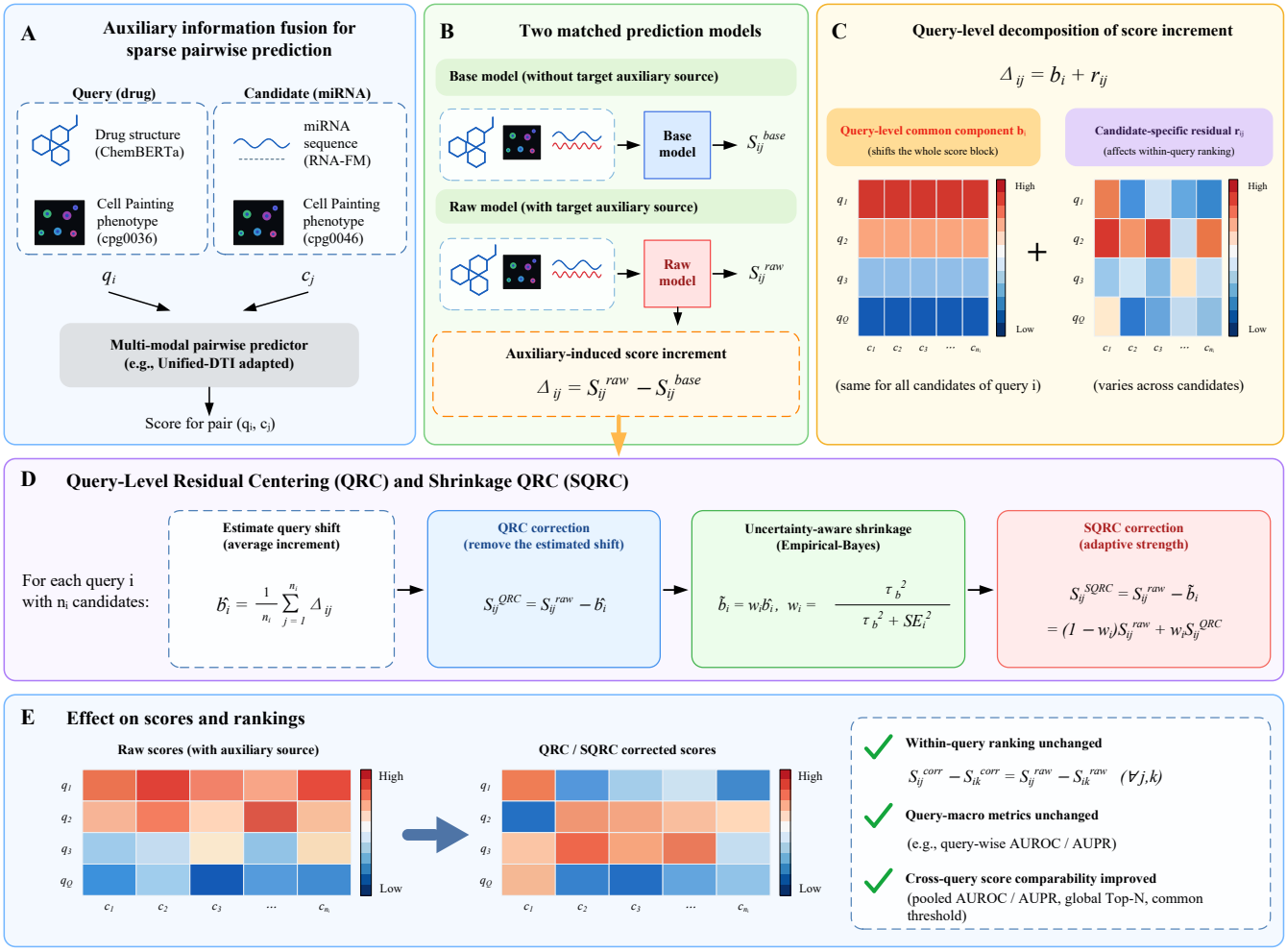


Fig. 1. QRC/SQRC workflow for sparse pairwise prediction. (A) Drug–miRNA queries and candidates with auxiliary representations. (B) Matched Base and Raw prediction branches, differing in the target auxiliary source. (C) Decomposition of the Raw-minus-Base score increment into a query-common component and a candidate-specific residual. (D) Query-level residual centering and uncertainty-aware shrinkage. (E) Effect of the correction on cross-query score placement while within-query candidate ordering is preserved.

label-independent query shift, and a zero-mean candidate residual:

$$S_{ij}^{\text{raw}} = S_{ij}^{\text{base}} + b_i^N + r_{ij}, \qquad \frac{1}{n_i}\sum_{j=1}^{n_i} r_{ij} = 0. \tag{6}$$

When the candidate mean estimates the nuisance shift exactly, QRC gives

$$\widehat{b}_i = b_i^N, \qquad S_{ij}^{\text{QRC}} = S_{ij}^{\text{base}} + r_{ij}. \tag{7}$$

Under these conditions the query-level nuisance is removed while the candidate-specific contribution is retained. The controlled experiments test this identity numerically and then relax it by introducing task-informative query components.

3.4. Shrinkage QRC

3.4.1. Estimation Uncertainty

Full centering treats every estimated query shift as equally reliable. That assumption is weak when a query has few candidates or when its auxiliary-induced increments vary widely. For query i, s_i^2 denotes the variance of Δ_{ij} across candidates, and SE_i^2 denotes the estimated variance of the query mean:

$$s_i^2 = \text{Var}_j(\Delta_{ij}), \qquad \text{SE}_i^2 = \frac{s_i^2}{n_i}. \tag{8}$$

The variance in (8) is computed across candidates for one fixed query. Let τ_b^2 denote the between-query variance of the latent shifts after accounting for sampling uncertainty. The observed dispersion of $\widehat{b}_i$ contains both genuine between-query heterogeneity and estimation noise, so we use the method-of-moments estimate

$$\widehat{\tau}_b^2 = \max\left\{\text{Var}_i(\widehat{b}_i) - \text{mean}_i(\text{SE}_i^2),\, 0\right\}. \tag{9}$$

The outer variance in (9) is taken over queries. This distinction is important: the within-query variance controls how uncertain one shift estimate is, whereas the across-query variance measures how strongly query-level shifts vary over the dataset after subtracting average estimation noise. The maximum with zero prevents a negative variance estimate when the observed dispersion can be explained by sampling noise alone.

3.4.2. Empirical-Bayes Shrinkage

A random-effects interpretation gives the shrinkage rule a probabilistic meaning. Here b_i is the latent query shift, $\widehat{b}_i$ is its observed estimate, and τ_b^2 and SE_i^2 are the between-query and within-query uncertainty terms. Bayesian molecular-property prediction has likewise examined informative priors under limited evidence [40]. We use the following random-effects model to regulate correction strength:

$$b_i \sim \mathcal{N}(0, \tau_b^2), \qquad \widehat{b}_i \mid b_i \sim \mathcal{N}(b_i, \text{SE}_i^2). \tag{10}$$

Under this Gaussian model, the posterior mean $\widetilde{b}_i$ of the latent shift is

$$\widetilde{b}_i = \mathbb{E}[b_i \mid \widehat{b}_i] = w_i \widehat{b}_i, \tag{11}$$

where w_i is the query-specific shrinkage weight

$$w_i = \frac{\tau_b^2}{\tau_b^2 + \text{SE}_i^2}. \tag{12}$$

The weight approaches one when the current query shift is estimated precisely relative to the global between-query variance. It approaches zero when the shift estimate is dominated by uncertainty. SQRC subtracts the shrunken estimate:

$$S_{ij}^{\text{SQRC}} = S_{ij}^{\text{raw}} - \widetilde{b}_i = S_{ij}^{\text{raw}} - w_i \widehat{b}_i. \tag{13}$$

An equivalent expression shows that SQRC lies between Raw and full QRC correction:

$$S_{ij}^{\text{SQRC}} = (1 - w_i) S_{ij}^{\text{raw}} + w_i S_{ij}^{\text{QRC}}. \tag{14}$$

This shrinkage addresses numerical uncertainty in estimating a query shift. It does not determine whether the latent shift carries useful label information. That question requires validation evidence or domain knowledge.

3.5. Identifiability and Applicability Conditions

A real query-level component can contain both nuisance and information. We write

$$b_i = b_i^N + b_i^I. \tag{15}$$

where the superscript N denotes a nuisance component and I an informative component. Unlabeled test scores do not generally identify these components separately. A numerically accurate estimate of the total shift can therefore remove useful signal when the informative component is large. The method is appropriate when matched Base and Raw predictions are available, each query has enough candidates for a stable estimate, the dominant distortion is approximately additive, and the deployment task cares about cross-query score comparability in addition to within-query ranking.

The same analysis gives clear cases in which correction has limited value. A task evaluated only with within-query Recall@K or NDCG@K will obtain the same ranking under QRC. If Raw already damages the ordering of candidates within a query, subtracting a common constant cannot repair that ordering. If query-level variation encodes a task-relevant property, full centering can discard useful information. The PRISM strict-cold and Davis DTI experiments are included specifically to test these two failure modes.

3.6. Algorithm and Computational Cost

Algorithm 1 summarizes the complete post-hoc procedure. The only pass over pair scores computes increments and query statistics. For P scored pairs and Q queries, the time complexity is O(P) and the additional memory requirement is O(Q) when sufficient statistics are accumulated per query.

Algorithm 1 Query-Level Residual Centering and Shrinkage Correction

Require: Matched Base and Raw scores, query groups, validation-query statistics

Ensure: QRC and SQRC scores, estimated shifts, standard errors, shrinkage weights

for each observed pair (i, j) **do**
  $\Delta_{ij} \leftarrow S_{ij}^{\text{raw}} - S_{ij}^{\text{base}}$
end for
for each query i **do**
  $n_i \leftarrow$ candidate count; $\widehat{b_i} \leftarrow \text{mean}_j\, \Delta_{ij}$
  $\text{SE}_i^2 \leftarrow \text{Var}_j(\Delta_{ij})/n_i$
  $S_{ij}^{\text{QRC}} \leftarrow S_{ij}^{\text{raw}} - \widehat{b_i}$ for every candidate j
end for
$\tau_b^2 \leftarrow \max\{\text{Var}_i(\widehat{b_i}) - \text{mean}_i(\text{SE}_i^2), 0\}$ on validation queries
for each query i **do**
  $w_i \leftarrow \tau_b^2/(\tau_b^2 + \text{SE}_i^2)$
  $S_{ij}^{\text{SQRC}} \leftarrow S_{ij}^{\text{raw}} - w_i\widehat{b_i}$ for every candidate j
end for
Report pooled and query-macro metrics and shift diagnostics

4. Results

4.1. Experimental Settings and Evaluation Protocols

The evaluation is organized as a sequence of increasingly realistic tests. Controlled interventions first examine additive recovery, ranking invariance, injected query bias, and informative query-level signal. The primary Cell Painting MDA study then tests the failure mode in biological screening. Frozen-task backbone changes probe architecture dependence; PRISM changes the auxiliary source; and Davis DTI tests the semantic boundary. Finally, counterfactual shuffling, candidate-set subsampling, SQRC stress tests, and post-hoc baselines isolate component roles, estimation stability, and differences from related corrections.

The primary MDA cohort contains 65 drugs, 1195 microRNAs, and 3443 known associations. Drug Cell Painting profiles come from the public Cell Painting Gallery dataset cpg0036-EU-OS-bioactives, and miRNA perturbation profiles come from cpg0046-microrna. Restricting both sources to HepG2 places the two phenotype views in the same cellular context. We use the released calibrated and aggregated profiles rather than raw images, apply feature-compatibility and numerical-quality checks, and retain 39 shared phenotype features for each side.

The primary predictor, denoted B0, is our MDA adaptation of the two-stage Unified-DTI multimodal contrastive framework. Its four frozen input views are drug structure encoded by ChemBERTa (384D), miRNA sequence encoded by RNA-FM (640D), drug Cell Painting phenotype (39D), and miRNA Cell Painting phenotype (39D). The Base condition uses the molecular branch without the target phenotype contribution, whereas Raw uses the matched four-view fusion. Random, miRNA-cold, drug-cold, and strict-cold protocols are evaluated over 25 seeds; drug-cold and strict-cold are the primary deployment settings because unseen drugs must be compared in one shared score space.

PRISM uses the 19Q4 Primary Screen. All 65 MDA drugs are mapped to PRISM, including 63 exact InChIKey matches; VELIPARIB and prednisolone are retained by exact standardized-name matching and are not described as structure-identical matches. After STR quality control and completeness filtering, 467 cell lines remain. Principal component analysis is fitted only on identity-clean external treatments to prevent downstream target drugs from influencing the phenotype basis. The resulting drug representation has 39 components with cumulative explained variance 0.7520. Each drug-cold seed contains 53 training drugs, 6 validation drugs, and 6 test drugs. In strict-cold evaluation, microRNAs are also separated and each test set contains 6 by 120 query-candidate pairs.

Davis DTI uses MMDG-DTI, a multimodal DTI model with language-model and structural branches, under a strict drug-cold protocol with 25 formal seeds numbered 5 through 29. Base is the model's dti1d language-model branch and Raw is the official multimodal fusion that incorporates the complementary dti2d structural branch. This experiment deliberately tests the semantic boundary of QRC because a drug-level score component may reflect global binding propensity rather than nuisance offset.

4.2. Metrics and Matched Comparisons

We report pooled AUROC and AUPR together with query-macro AUROC and AUPR. Pooled metrics flatten all query-candidate pairs and therefore include cross-query score comparisons, whereas query-macro metrics are computed separately inside each query and then averaged with equal query weight. Key real-data comparisons are reported as mean ± standard deviation across 25 seeds. Their separation is diagnostic: improved query-macro performance together with degraded pooled performance indicates that within-query discrimination and cross-query score placement have moved in different directions.

The main comparisons are Base, Raw, QRC, and SQRC. Formal post-hoc controls include RawCenter, RawZScore, a median-residual correction, and global Platt calibration of Raw scores. These controls distinguish query-location correction from generic within-query normalization, robust residual estimation, and probability calibration. Mechanism diagnostics include the mean absolute Raw-Base score change before and after correction, the query-shift variance ratio that measures how much auxiliary-induced variation is concentrated in query means, and seed-wise query-macro invariance. Brier score, 10-bin expected calibration error, and negative log-likelihood are reported only for methods that return calibrated probabilities.

4.3. Controlled Query-Level Shift Experiments

The additive ground-truth experiment uses 25 seeds. The residual-strength parameter α takes values $\{0, 0.5, 1, 1.5\}$, and the nuisance-shift parameter β takes values $\{0, 0.25, 0.5, 1, 2\}$. Because both components are known, recovery can be checked

Table 1
Experimental Settings and Their Roles

Setting	Data and Protocol	Primary Purpose
Controlled additive	25 seeds; α = 0, 0.5, 1, 1.5; β = 0, 0.25, 0.5, 1, 2	Recovery and invariance
Cell Painting MDA	Unified-DTI-adapted four-view MDA; cpg0036/cpg0046 HepG2; 65×1195; 25 seeds	Functional phenotype evaluation
Cross-backbone MDA	Same CP MDA drug-cold; HyperAttentionDTI-derived, DLST-MDA, HGGN; 25 seeds	Backbone portability
PRISM MDA	65 drugs x 1195 miRNAs; 39D PRISM representation; 25 seeds	Independent auxiliary source
Davis DTI	MMDG-DTI; Davis drug-cold; 25 formal seeds	Cross-task boundary test

against the exact nuisance-free target. QRC recovers that target with numerical error on the order of 10^{-19}, while query-macro AUROC and AUPR remain unchanged. This test verifies the additive identity underlying the correction rather than relying on a favorable real-data split.

A separate query-bias injection test starts from the real cold-split score matrices and adds an increasingly strong common shift to each query, using multipliers {0, 0.25, 0.5, 1, 2, 4}. For drug-cold evaluation, Raw pooled AUROC decreases from 0.5359 at zero injection to 0.5253 at the largest multiplier, whereas QRC remains at 0.5893 and macro AUROC remains 0.6923. The miRNA-cold setting is more sensitive: Raw pooled AUROC decreases from 0.9314 to 0.6250 across the same multipliers, while QRC remains 0.9307 and macro AUROC remains 0.9441. The maximum QRC deviation is at numerical precision, directly confirming that a common additive query shift changes cross-query placement without changing query-internal ordering.

As shown in Fig. 2(C) and Fig. 2(D), the informative-shift study adds both a nuisance query component and a task-informative query component. Their strengths are controlled by β and γ, respectively, with $\beta, \gamma \in \{0, 0.25, 0.5, 1, 2\}$. Across 25 seeds, the 25 β-γ conditions yield 625 valid cases. The phase boundary is exact: QRC gives the lower recovery error in all 250 cases with $\beta > \gamma$, Raw is better in all 250 cases with $\beta < \gamma$, and the 125 $\beta = \gamma$ cases are tied. Raw recovery RMSE follows β and QRC recovery RMSE follows γ to numerical precision. QRC therefore removes the observable query-level location component whether it is nuisance or informative; correction is beneficial only when the nuisance contribution dominates.

4.4. Primary Evaluation on Cell Painting MDA

Cell Painting produces a split-dependent pattern, as shown in Fig. 2(A) and Fig. 2(B). Under random evaluation, Base achieves 0.9286 ± 0.0057 AUROC and 0.1358 ± 0.0248 AUPR; Raw improves these values to 0.9350 ± 0.0055 and 0.1472 ± 0.0274, while QRC gives 0.9327 ± 0.0051 and 0.1363 ± 0.0255. Under miRNA-cold evaluation, Base obtains 0.9233 ± 0.0072 AUROC and 0.3394 ± 0.0191 AUPR, Raw reaches 0.9314 ± 0.0059 and 0.3649 ± 0.0222, and QRC gives 0.9307 ± 0.0060 and 0.3681 ± 0.0254. These settings show that query-level movement is not uniformly harmful.

The drug-side cold settings expose the deployment failure more clearly. In drug-cold evaluation, Base obtains 0.5828 ± 0.1501 AUROC and 0.0577 ± 0.0429 AUPR. Adding Cell Painting directly reduces Raw to 0.5359 ± 0.1609 and 0.0449 ± 0.0361, whereas QRC restores 0.5893 ± 0.1591 and 0.0641 ± 0.0537; SQRC is nearly identical. Under strict-cold evaluation, Base is 0.5401 ± 0.1721 and 0.0626 ± 0.0512, Raw decreases to 0.4964 ± 0.1609 and 0.0499 ± 0.0420, and QRC restores 0.5438 ± 0.1806 and 0.0698 ± 0.0618. Because QRC leaves each drug-specific miRNA ordering unchanged, the recovery comes from realigning unseen-drug score blocks rather than re-learning candidate ranking.

The practical interpretation is direct. In a shared screening campaign, a common score shift can cause one unseen drug to dominate a global candidate list even when its internal miRNA ranking is reasonable. QRC removes only this auxiliary-induced location change, so the useful candidate-specific component is retained. The opposite behavior under random and miRNA-cold evaluation also shows why QRC should be treated as a conditional reliability correction rather than an always-on performance booster.

4.5. Cross-Backbone Validation under Fixed MDA Splits

To test whether the score-placement phenomenon depends on B0, we keep the MDA cohort, Cell Painting source, drug-cold splits, seeds, and candidate universe fixed and change the upstream predictor. We evaluate a HyperAttentionDTI-derived sequence-attention backbone, DLST-MDA, and HGGN. Base and Raw remain matched within each backbone; QRC and SQRC are applied after score generation. Original model provenance is recorded in the accompanying source ledger.

As shown in Fig. 3(D), Raw lowers the mean pooled metrics relative to the corresponding Base for all three additional backbones, whereas QRC restores the pooled scores toward the Base-level cross-drug placement. HyperAttentionDTI-derived Raw-to-QRC changes from 0.5574 ± 0.1475 to 0.6108 ± 0.1630 AUROC and from 0.0543 ± 0.0398 to 0.0807 ± 0.0769 AUPR. DLST-MDA changes from 0.5903 ± 0.1504 to 0.6425 ± 0.1238 AUROC and from 0.0635 ± 0.0407 to 0.0732 ± 0.0439 AUPR. HGGN changes from 0.5729 ± 0.1708 to 0.6237 ±

0.1283 AUROC and from 0.0597 ± 0.0411 to 0.0652 ± 0.0494 AUPR. For HGGN, the representation graph is built from training positives only; validation and test MDA edges and all test labels are excluded. Recurrence across sequence, sequence-structure, and heterogeneous-graph predictors shows that the score-placement effect is not tied to one architecture. The substantial seed-wise variability, however, argues against interpreting QRC as a universal accuracy-improvement module.

4.6. Validation with an Independent PRISM Auxiliary Source

As shown in Fig. 3(C), PRISM drug-cold provides the clearest separation between within-query discrimination and cross-query comparability. Base pooled AUROC and AUPR are 0.5828 ± 0.1501 and 0.0577 ± 0.0429. Raw decreases these values to 0.5545 ± 0.1665 and 0.0553 ± 0.0441. At the same time, query-macro AUROC increases from 0.6326 ± 0.1016 to 0.6842 ± 0.0751, and query-macro AUPR increases from 0.0685 ± 0.0436 to 0.0843 ± 0.0516. The auxiliary phenotype therefore improves average single-drug microRNA ranking while moving different drug score blocks to less compatible

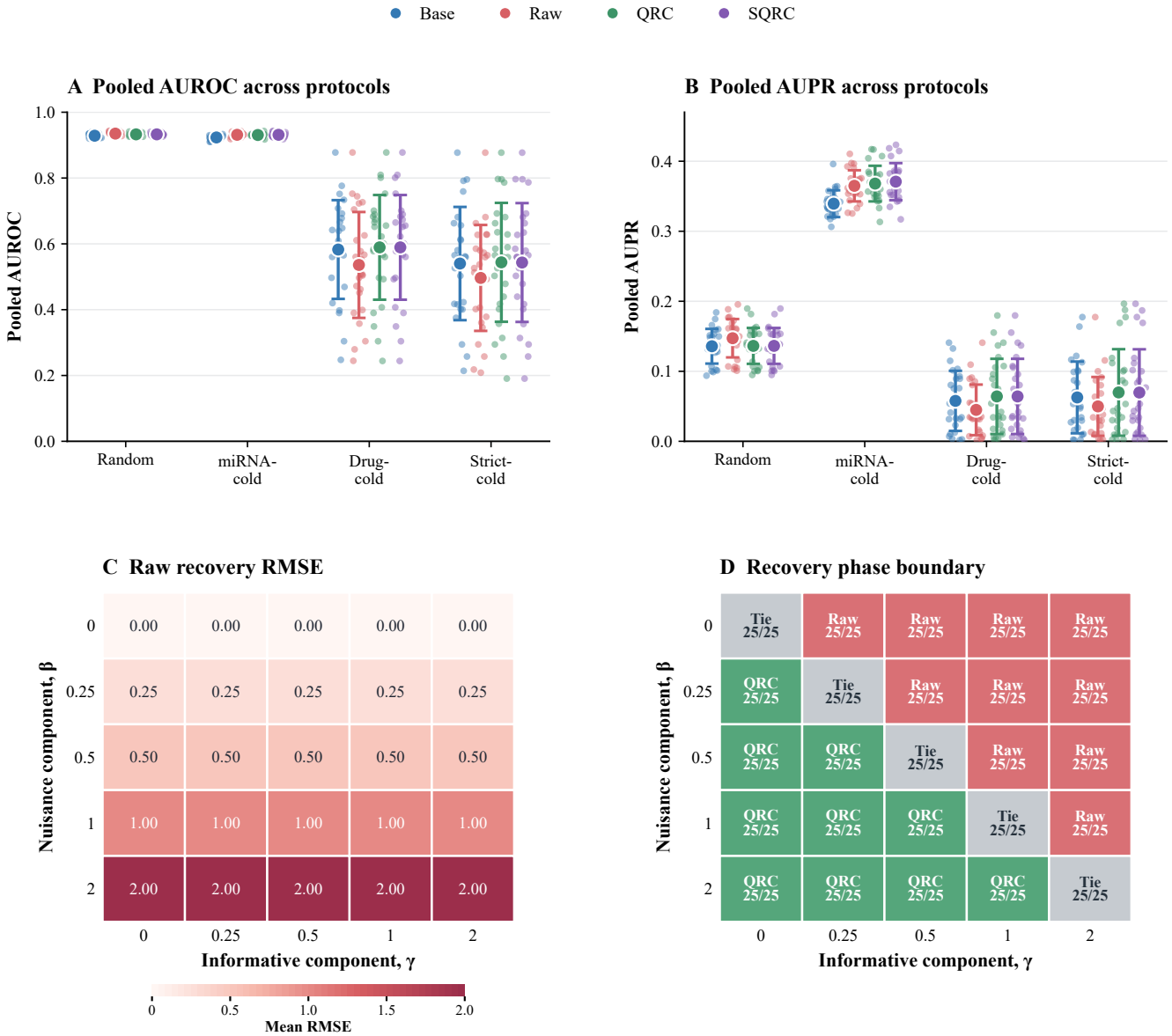


Fig. 2. Pooled screening performance and controlled recovery under query-level shifts. (A) Pooled AUROC and (B) pooled AUPR across random, miRNA-cold, drug-cold, and strict-cold protocols. Small points represent 25 seeds per method and protocol; large points and bars show mean ± SD. (C) Mean Raw recovery RMSE across nuisance strength β and informative strength γ, based on 25 seeds per condition. (D) Method with the lower recovery error under the same conditions; cells report wins or ties out of 25 seeds. The 625 cases are controlled computational experiments, not biological samples.

Table 2
Cell Painting MDA Drug-Cold Results (25 Seeds, Mean ± SD)

Split	Method	AUROC (mean ± SD)	AUPR (mean ± SD)
Drug-cold	Base	0.5828 ± 0.1501	0.0577 ± 0.0429
	Raw	0.5359 ± 0.1609	0.0449 ± 0.0361
	QRC	0.5893 ± 0.1591	0.0641 ± 0.0537
	SQRC	0.5893 ± 0.1591	0.0641 ± 0.0537
Strict-cold	Base	0.5401 ± 0.1721	0.0626 ± 0.0512
	Raw	0.4964 ± 0.1609	0.0499 ± 0.0420
	QRC	0.5438 ± 0.1806	0.0698 ± 0.0618
	SQRC	0.5433 ± 0.1805	0.0696 ± 0.0618

locations.

QRC preserves the Raw macro values exactly while restoring pooled AUROC to 0.5781 ± 0.1498 and pooled AUPR to 0.0625 ± 0.0447. The mean absolute Raw-Base score difference is 0.1816 ± 0.0573 and falls to 0.0868 ± 0.0055 after QRC, a reduction of approximately 52.2%. The query-shift variance ratio is 0.6707 ± 0.1697, showing that a substantial fraction of the auxiliary-induced score change is concentrated in the common query component. All 25 seeds satisfy exact query-macro invariance. These diagnostics are summarized in Fig. 3(A).

Across 25 seeds, QRC reduces the mean pooled degradation introduced by Raw without changing Raw query-macro values. The recovery is not identical in every split: under drug-cold, QRC returns close to the Base AUROC while slightly exceeding Base AUPR, whereas strict-cold remains below Base AUROC because the within-query ranking itself has already deteriorated. This distinction supports the score-decomposition mechanism without requiring a claim of uniform predictive superiority. In the same PRISM drug-cold test pools, the fixed Top-5% budget retains 3,983 Raw-selected seed–candidate memberships and exchanges 4,992; mean known-positive hits per seed rise from 18.28 under Raw to 20.64 under QRC (Fig. 3(B)). These counts do not establish statistical significance.

PRISM strict-cold exposes a different limitation. Raw query-macro AUROC falls from 0.5924 to 0.5311, so within-query ranking itself deteriorates. QRC raises pooled AUROC only from 0.5012 to 0.5135, still below the Base value of 0.5401. Pooled AUPR moves from 0.0489 to 0.0613, close to the Base value of 0.0626. The query-shift variance ratio remains approximately 0.6663 and the mean absolute Raw-Base difference is reduced by about 50.8%, yet score-location correction cannot recreate candidate-specific signal that has already been lost.

4.7. Cross-Task Evaluation on Davis DTI

As shown in Fig. 4(A), Davis drug-cold tests whether the same query-level phenomenon should always be interpreted as nuisance using MMDG-DTI. Across 25 formal seeds, its dti1d Base branch obtains 0.6993 ± 0.0702 AUROC and 0.4936 ± 0.0901 AUPR. The official multimodal Raw output improves to 0.7208 ± 0.0570 and 0.5116 ± 0.0881. QRC gives 0.7038 ± 0.0730 and 0.4968 ± 0.0921, while SQRC gives 0.7039 ± 0.0728 and 0.4970 ± 0.0920. Here, retaining the query-level component is beneficial, consistent with the possibility that drug-level binding propensity or another global query property carries task information.

The controlled informative-shift experiment and Davis result establish the semantic boundary of the method. QRC can esti-

Table 3
Cross-Backbone Drug-Cold Results (25 Seeds, Mean ± SD)

Backbone	Method	Pooled AUROC	Pooled AUPR
HyperAttn.-DTI	Base	0.6327 ± 0.1389	0.0709 ± 0.0565
	Raw	0.5574 ± 0.1475	0.0543 ± 0.0398
	QRC	0.6108 ± 0.1630	0.0807 ± 0.0769
	SQRC	0.6108 ± 0.1630	0.0806 ± 0.0768
DLST-MDA	Base	0.6427 ± 0.1258	0.0704 ± 0.0449
	Raw	0.5903 ± 0.1504	0.0635 ± 0.0407
	QRC	0.6425 ± 0.1238	0.0732 ± 0.0439
	SQRC	0.6422 ± 0.1240	0.0731 ± 0.0435
HGGN	Base	0.6047 ± 0.1339	0.0681 ± 0.0462
	Raw	0.5729 ± 0.1708	0.0597 ± 0.0411
	QRC	0.6237 ± 0.1283	0.0652 ± 0.0494
	SQRC	0.6169 ± 0.1396	0.0651 ± 0.0492

Table 4
PRISM Pooled and Query-Macro Results

Setting	Method	Pooled		Query-macro	
		AUROC	AUPR	AUROC	AUPR
Drug-cold	Base	0.5828	0.0577	0.6326	0.0685
	Raw	0.5545	0.0553	0.6842	0.0843
	QRC	0.5781	0.0625	0.6842	0.0843
	SQRC	0.5781	0.0625	0.6842	0.0843
Strict-cold	Base	0.5401	0.0626	0.5924	0.1358
	Raw	0.5012	0.0489	0.5311	0.1250
	QRC	0.5135	0.0613	0.5311	0.1250
	SQRC	0.5135	0.0612	0.5311	0.1250

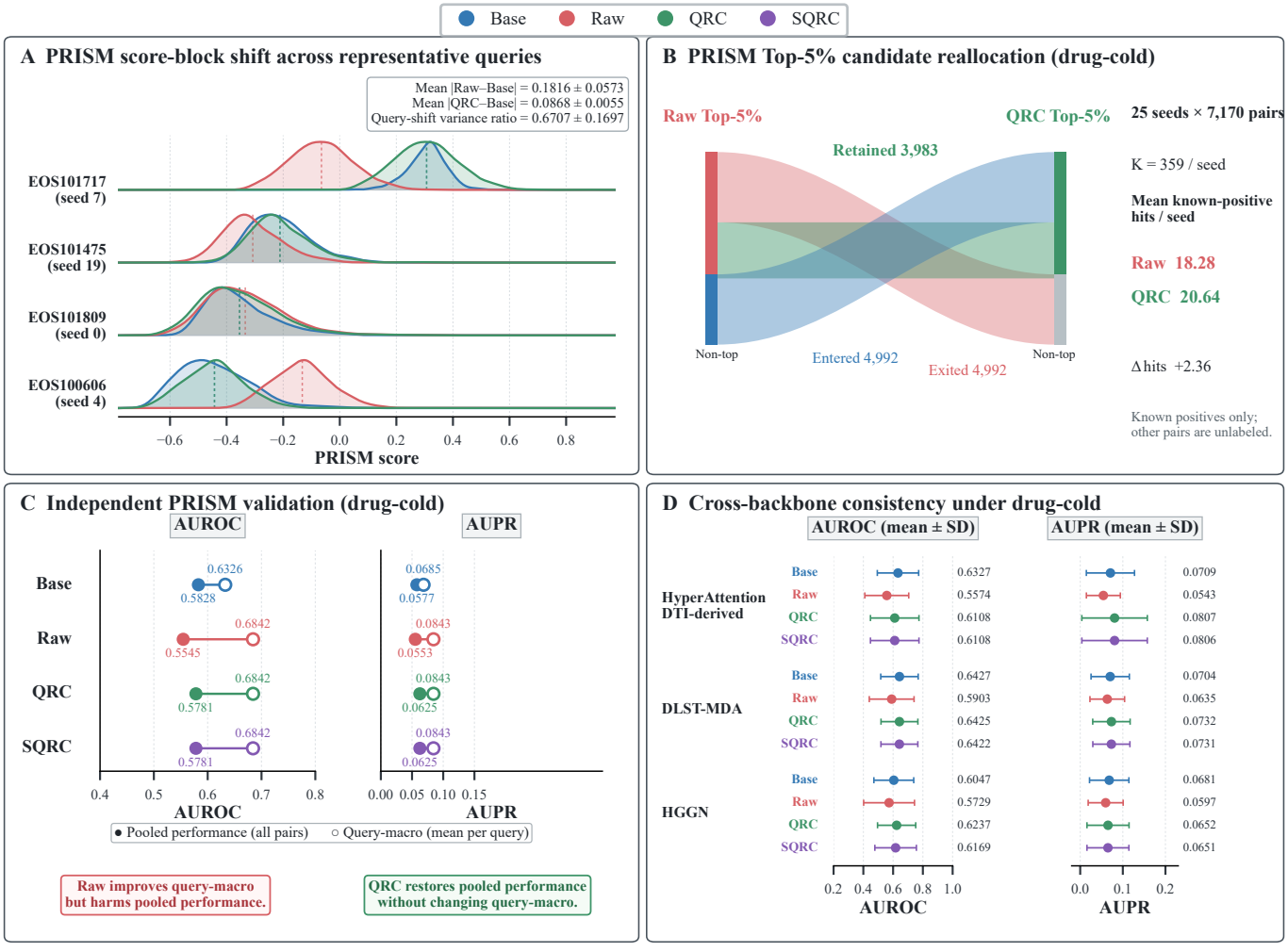


Fig. 3. Query-level score shifts, candidate reallocation, and independent validation. (A) Base, Raw, and QRC score densities for four real PRISM drug-cold seed–drug queries; dashed lines mark means, and the statistics box summarizes all 25 seeds. (B) Raw-to-QRC Top-5% candidate migration within the same PRISM drug-cold test pool (7,170 pairs and K = 359 per seed). Ribbon widths reflect aggregate seed–candidate counts; hit counts are known positives per seed, while other pairs remain unlabeled. (C) Pooled and query-macro AUROC/AUPR across 25 PRISM drug-cold seeds, shown by filled and open circles. (D) Cross-backbone drug-cold means ± SD across 25 seeds for each method.

mate and remove a common additive component, but the unlabeled score matrix alone cannot determine whether that component is nuisance or signal. Deployment should therefore combine matched Base and Raw behavior, the separation between macro and pooled metrics, controlled perturbation evidence, and validation-set performance before enabling full correction.

4.8. Component Permutation and Mechanism Analysis

Counterfactual interventions isolate the query-common and candidate-specific terms in (2). Under drug-cold evaluation, Raw and QRC pooled AUROC are 0.5359 and 0.5893. Shuffling the query component b_i across queries changes pooled AUROC to 0.5772 while leaving macro AUROC at 0.6923, exactly the Raw/QRC macro value. Shuffling r_{ij} within each query instead lowers macro AUROC to 0.6171 and pooled AUROC to 0.5275. The same separation appears under miRNA-cold evaluation: bias shuffling gives pooled AUROC 0.9288 with macro AUROC fixed at 0.9441, whereas residual shuffling lowers them to 0.9078 and 0.9213. The AUPR results follow the same pattern: 0.0634 versus 0.0465 in drug-cold and 0.3570 versus 0.2904 in miRNA-cold for bias and residual shuffling, respectively. These interventions support the statistical distinction between query-common location changes and candidate-specific ranking changes; they do not establish causal biological roles. The counterfactual results are shown in Fig. 4(B).

Mechanism diagnostics are reported separately from predictive accuracy. In PRISM drug-cold, the mean absolute Raw-Base score change is 0.1816 ± 0.0573 and falls to 0.0868 ± 0.0055 after QRC, a 52.2% reduction; the query-shift variance ratio is 0.6707 ± 0.1697, and all 25 seeds satisfy exact macro invariance. PRISM strict-cold shows a similar shift ratio of approximately 0.6663 and a 50.8% reduction in the mean absolute score change, yet QRC cannot recover the lost within-query ranking. Davis drug-cold has a smaller query-shift ratio, 0.4471 ± 0.1533, while Raw remains better than QRC in pooled AUROC. Shift magnitude therefore diagnoses how much of the auxiliary change is query-common, but it does not determine whether that component should be removed.

4.9. Candidate-Set Sensitivity and SQRC Stress Tests

As shown in Fig. 4(C), candidate-set subsampling measures how strongly the estimated query shift depends on which candidates are available. Across 25 seeds, drug-cold QRC is already stable with 10% of candidates: relative to full-candidate QRC, pooled AUROC changes by -0.000119 ± 0.000678 and AUPR by -0.000058 ± 0.000328, with a bias-estimate MAE of 0.004003. In miRNA-cold evaluation, the 10% subset is less stable, with AUROC and AUPR changes of -0.003078 ± 0.001665 and -0.018339 ± 0.010811 and a bias MAE of 0.025496. At 50% candidates these differences contract to -0.000369 ± 0.000215 AUROC, -0.002609 ± 0.001825 AUPR, and 0.008259 bias MAE; at 75% they fall to -0.000155 ± 0.000101, -0.000831 ± 0.000726, and 0.004649. The experiment exposes a finite-candidate estimation error rather than an instability of the correction rule itself.

As shown in Fig. 4(D), the SQRC stress test isolates the same issue under controlled noise. It contains 800 cases with candidate counts from 5 to 1195 and increment-noise standard deviation from 0.25 to 2.0. Uncertainty is almost perfectly aligned with shrinkage strength: Spearman correlation is 0.999725 between uncertainty and $1 - w$, and 0.919424 between uncertainty and the MSE gain of SQRC over QRC. SQRC has lower bias-estimation MSE in 563 of 800 cases. In the most difficult reported condition, $\sigma = 2$ and $n = 5$, the weight falls to 0.598001 and the bias MSE decreases from 0.838862 under QRC to 0.525022 under SQRC, a 37.04% reduction. With broad candidate support, w approaches one; PRISM ($w \approx 0.999$) consequently yields nearly identical QRC and SQRC corrections.

4.10. Post-hoc Baseline Comparison

The 125-case post-hoc study compares QRC with methods that target related but different output properties. Direct centering and z-scoring can improve pooled AUROC in the drug-side cold settings, but they also remove query-level structure that was already present before auxiliary fusion. This cost is visible in settings where that structure remains useful: under random evaluation, RawCenter falls to 0.632570 AUROC and 0.013274 AUPR, compared with 0.932725 and 0.136302 for QRC; under miRNA-cold evaluation, the corresponding values are 0.917712 and 0.313710 for RawCenter versus 0.930738 and 0.368056 for QRC. The formal drug-cold and strict-cold comparison is summarized in Table VI. No method dominates every metric, which is consistent with their different estimands.

Probability calibration also improves probability quality, and recent calibration studies clarify both biased-label and high-dimensional settings. In the reported MDA drug-cold calibration experiment, Brier score, 10-bin ECE, and NLL change from 0.2454, 0.4504, and 0.6838 for Raw to 0.0445, 0.0478, and 0.2137 after Platt calibration; under strict-cold they change from 0.2400, 0.4432, and 0.6728 to 0.0467, 0.0525, and 0.2177. Because a single strictly increasing sigmoid applied globally to one fixed score matrix preserves pair ordering, its pooled AUROC and AUPR are identical to Raw in Table VI. The probability gains therefore reflect improved numerical calibration rather than improved ranking.

5. Discussion

The experiments reveal an important distinction between predictive discrimination and score comparability in sparse pairwise learning. Auxiliary information can improve candidate ranking within individual queries while changing the relative positions of their score distributions. The PRISM results provide a clear example of this phenomenon, where improved query-macro discrimination is accompanied by reduced pooled performance. Such discrepancies are particularly relevant to applications that require candidates from different queries to compete under a shared screening budget or a common decision threshold. Evaluating prediction models solely through their ability to distinguish candidates within individual queries may therefore overlook an important aspect of decision reliability.

QRC addresses this problem by estimating the common score change introduced by auxiliary information through matched Base and Raw predictions. Rather than directly centering the original prediction scores, QRC operates on their increments and removes the estimated query-level component. This formulation preserves the score location information already captured by Base while retaining the candidate-specific changes introduced by Raw. The correction also leaves all within-query score differences unchanged, providing an exact ranking invariance property without modifying the underlying prediction

Table 5
Davis DTI Drug-Cold Cross-Task Results (25 Seeds, Mean ± SD)

Method	AUROC (mean ± SD)	AUPR (mean ± SD)
Base	0.6993 ± 0.0702	0.4936 ± 0.0901
Raw	0.7208 ± 0.0570	0.5116 ± 0.0881
QRC	0.7038 ± 0.0730	0.4968 ± 0.0921
SQRC	0.7039 ± 0.0728	0.4970 ± 0.0920

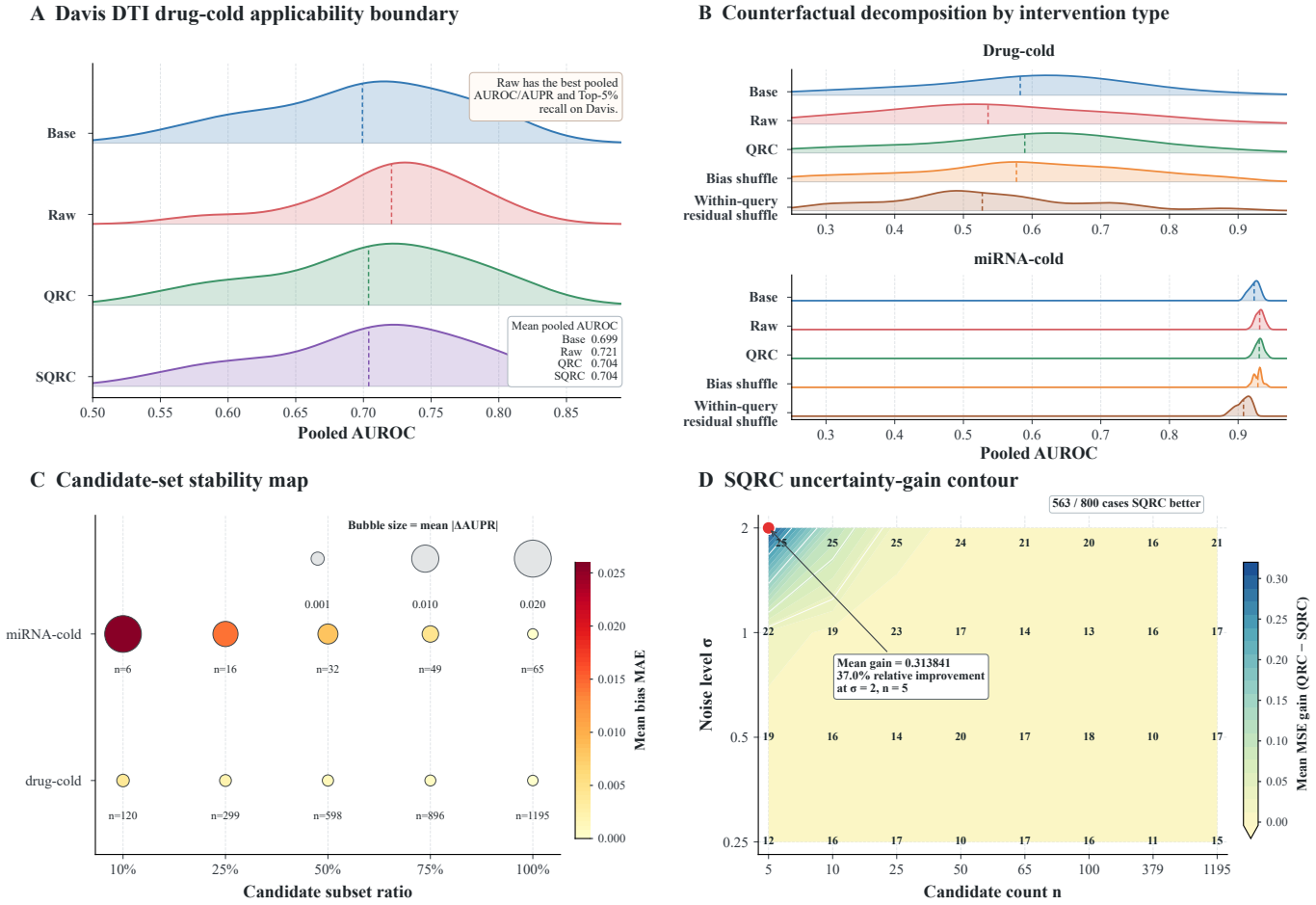


Fig. 4. Applicability boundary, counterfactual decomposition, candidate-set stability, and SQRC uncertainty benefit. (A) Pooled-AUROC distributions over 25 paired Davis DTI drug-cold seeds. (B) Pooled-AUROC distributions for Base, Raw, QRC, bias shuffling, and within-query residual shuffling under drug-cold and miRNA-cold protocols; each shuffle condition comprises 25 seeds × 20 repeats, not 500 independent seeds. (C) Candidate-set stability; color encodes mean bias MAE, bubble area mean absolute ΔAUPR, and labels give candidates per query. (D) Mean QRC-minus-SQRC bias MSE gain at 32 measured noise–candidate conditions (25 seeds each); labels give SQRC wins out of 25. Contours between measured conditions are visual interpolation.

Table 6
Post-hoc Baseline Comparison Under Drug-Cold Splits

Method	Drug AUROC	Drug AUPR	Strict AUROC	Strict AUPR
Base	0.5828	0.0577	0.5401	0.0626
Raw	0.5359	0.0449	0.4964	0.0499
Global Platt	0.5359	0.0449	0.4964	0.0499
MedianResidual	0.5894	0.0642	0.5438	0.0701
RawCenter	0.6045	0.0623	0.5855	0.0606
RawZScore	0.6045	0.0600	0.5831	0.0602
QRC	0.5893	0.0641	0.5438	0.0698
SQRC	0.5893	0.0641	0.5433	0.0696

model. The controlled experiments confirm the additive recovery behavior, while evaluations across different prediction backbones and auxiliary sources demonstrate that the observed score placement problem is not restricted to a particular model architecture or feature representation.

The effectiveness of correction nevertheless depends on the information contained in the query-level component. In PRISM strict-cold evaluation, auxiliary fusion reduces discrimination within individual queries. Since QRC only adjusts the common score location, it cannot recover candidate ranking information that has already been lost. The Davis DTI experiment reveals another limitation. The multimodal Raw predictor performs better when the query-level component is retained, suggesting that such changes may contain useful predictive information. This observation agrees with the controlled informative-shift experiments, where removing a task-relevant component increases recovery error. Although QRC can estimate the common shift without labeled test data, the distinction between nuisance and informative components cannot be established from the score distribution alone. Validation evidence remains necessary when deciding whether correction is appropriate.

SQRC extends the correction framework by accounting for uncertainty in query-level shift estimation. Its role becomes more relevant when candidate sets are small or score increments are highly variable. In the controlled stress tests, SQRC achieves lower estimation error in 563 of 800 cases, indicating that shrinkage can improve the reliability of the estimated correction under uncertain conditions. However, a more accurate estimate does not necessarily guarantee better predictive performance. The current formulation relies on assumptions about query-level variation and the uncertainty of candidate score increments. Dependencies among candidates and changes in candidate-set composition may affect these estimates. The similar performance of QRC and SQRC in settings with sufficiently large candidate sets also indicates that shrinkage provides limited additional benefit when estimation uncertainty is already small.

The practical significance of the proposed methods depends on how prediction scores are used. When the objective is restricted to ranking candidates separately within each query, QRC and SQRC do not change the resulting order. Their value becomes more apparent when predictions must be compared across queries for global candidate selection. The results suggest that pooled and query-macro metrics should be considered together when evaluating auxiliary information fusion, particularly under cold-start conditions. The choice of correction strategy should also account for validation performance and the stability of estimated query shifts.

This study identifies cross-query score comparability as an important reliability consideration in sparse pairwise learning and develops a lightweight framework for correcting auxiliary-induced score shifts. The theoretical analysis establishes the properties of the correction, while the experimental results demonstrate its effectiveness and clarify the conditions under which it should be applied. Future work will investigate validation-guided correction strategies that determine when and how strongly to adjust query-level scores. Extending the current approach to account for dependencies among candidates and more general forms of score distortion may further improve the reliability of machine learning systems used for joint decision-making.

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CRediT authorship contribution statement

Yun Zhou: Conceptualization, Supervision, Writing - review and editing; Min Zhang: Conceptualization, Methodology, Software, Validation, Formal analysis, In vestigation, Visualization, Writing - original draft; Maoqi Lian: Data curation, Investigation, Writing - review and editing; Wei Wang: Resources, Supervision, Writing - review and editing; Dong Liu: Resources, Supervision, Writing - review and editing;

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

https://github.com/HNUBIGroup/
QRC-SQRC-Reliable-Cross-Query-Scoring

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